Deterministic Compilation of Markdown Manuscripts into Compliant Word Documents

Preparing document submission formats programmatically is often undermined by non-deterministic large language models. Here we demonstrate a declarative compiler that maps standard Markdown to Word (.docx) documents that follow a declared format, with byte-identical reproducibility and pre- and post-export quality gates.

Document compilation; Agent Skills; Deterministic typography; Native OMML; Quality gates

1. Introduction

Scientific manuscripts require rigorous typographic consistency. Equations must align centrally with numbering pinned hard to the right margin, while tables demand standard three-line borders with repeating header rows across page splits.

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r},t)\right]\Psi(\mathbf{r},t) \quad \Leftrightarrow \quad \mathcal{H} = \sum_k \varepsilon_k\, c_k^\dagger c_k + \frac{1}{2}\sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j^\dagger c_l c_k \tag{1}$$

Table 1 Where each declared setting ends up

Aspect	Declared in `format.txt`	Checked after export
Typography	font, size, spacing, indent per style	yes
Equations	numbering and cross-references	yes
Tables	three-line or grid, repeating header	yes

Note: this table describes the declared format, not benchmark results.

2. Conclusion

Decoupling style contracts from manuscript content bridges the last mile of autonomous AI delivery.

References

[1] Knuth D E. The TeXbook[M]. Addison-Wesley, 1984.

[2] Lamport L. LaTeX: a document preparation system[M]. Addison-Wesley, 1994.